

CENTRALIZED AUDIT TRAIL COMPLIANCE ENGINE

Software Engineering — Problem Statement #50 Developer Tools & IT Operations

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PROBLEM STATEMENT:

The Centralized Audit Trail Compliance Engine is a compliance logging platform that ingests enterprise application event logs, identifies and masks sensitive Personally Identifiable Information (PII) such as Social Security Numbers and credit card numbers, and generates cryptographically verifiable audit records and exports. The system supports compliance officers and security auditors in securely searching, reviewing, verifying, and exporting audit information.

Actors and Use Cases

Primary Actors

- Compliance Officer
- Security Auditor

Supporting Actor

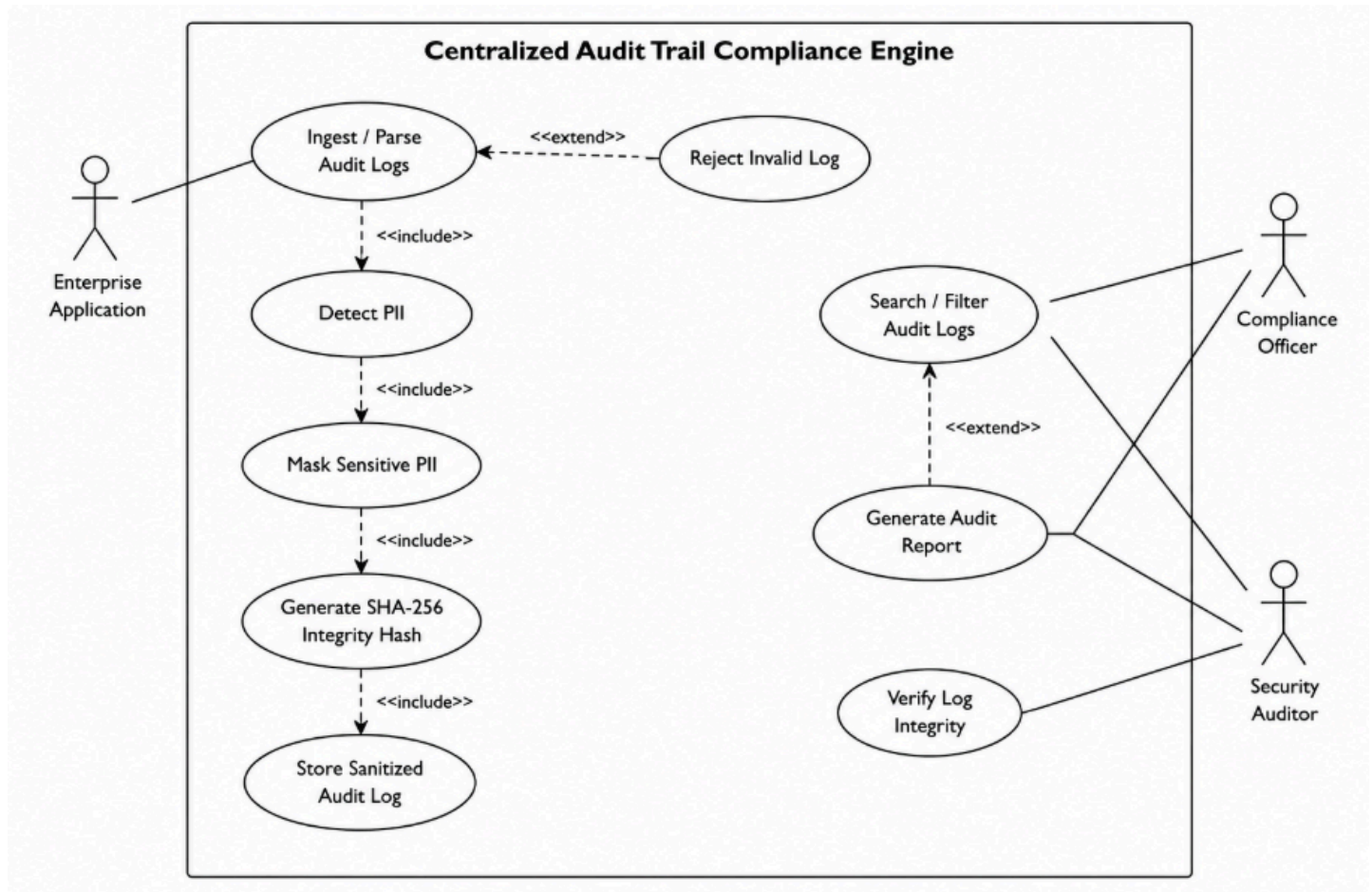
- Enterprise Application

REQUIREMENTS TABLE

ID	TYPE	DESCRIPTION	PRIORITY	ACCEPTANCE CRITERIA	RATIONALE
FR-001	Functional	The system shall parse structured JSON audit logs received from enterprise applications.	High	Pass: A valid JSON audit log is successfully parsed and its required event fields are extracted. Fail: Malformed or unsupported JSON is rejected and an error is recorded.	Structured log ingestion is required before the system can identify, process, and store audit events.
FR-002	Functional	The system shall identify and mask sensitive PII fields, including Social Security Numbers and credit card numbers, before storing audit logs.	High	Pass: Detected PII is replaced with masked values and no cleartext PII remains in the stored record. Fail: Any detected sensitive value is stored in cleartext.	Protects sensitive personal information and supports GDPR compliance.
FR-003	Functional	The system shall generate and store a SHA-256 integrity hash for every sanitized audit log.	High	Pass: A valid SHA-256 hash is generated and associated with the sanitized audit record. Fail: A record is stored without its required integrity hash.	Enables verification that audit records have not been altered after storage.
FR-004	Functional	The system shall allow authorized Compliance Officers and Security Auditors to search	High	Pass: An authorized user can retrieve audit records using supported filters. Fail: An unauthorized user	Enables efficient compliance reviews, security

		and filter historical audit logs.		can access audit records or search functionality.	investigations, and audit activities.
FR-005	Functional	The system shall generate exportable audit reports containing sanitized audit records and their integrity hashes.	Medium	Pass: Selected audit records can be exported without cleartext PII and include their SHA-256 hashes. Fail: The export contains unmasked PII or missing integrity information.	Compliance Officers and Security Auditors require verifiable records for audits and investigations.
NFR-001	Non-Functional — Performance	Audit log query endpoints shall support complex filtering across at least 10 million historical records with a response time of less than 1 second under the specified benchmark workload.	High	Pass: Benchmark testing demonstrates query response time below 1 second for the defined workload. Fail: The target response time is exceeded.	Large-scale audit investigations require fast retrieval of historical records.
NFR-002	Non-Functional — Security	The system shall enforce authenticated, role-based access to audit logs and prevent storage or export of unmasked sensitive PII.	High	Pass: Only authorized users can access audit records and security testing confirms that cleartext PII cannot be retrieved from storage or exports. Fail: Unauthorized access or cleartext PII exposure occurs.	Audit logs can contain sensitive information and must be protected from unauthorized access and data exposure.

USE CASE DIAGRAM



Use-Case Flow Specification

Field	Details
Use Case ID	UC-01
Use Case Name	Ingest and Sanitize Audit Log
Primary Actor	Enterprise Application
Supporting Actors	Audit Storage System
Priority	High
Goal	To receive an enterprise application audit log, identify and mask sensitive PII, generate a SHA-256 integrity hash, and securely store the sanitized log.

Preconditions

1. The Centralized Audit Trail Compliance Engine is running and available.
2. The Enterprise Application is authorized to submit audit logs.
3. The audit log is provided in the supported structured JSON format.
4. The audit storage system is available.

Postconditions

Success:

1. The audit log has been successfully validated.
2. Sensitive PII has been identified and masked.
3. A SHA-256 integrity hash has been generated.
4. The sanitized audit log and its hash have been securely stored.
5. Cleartext PII has not been stored.

Failure:

1. The invalid or unprocessable audit log is rejected.
2. The processing error is recorded.
3. No unsanitized audit log containing cleartext PII is stored.

Main Success Scenario

1. The Enterprise Application sends a structured JSON audit log to the Compliance Engine.
2. The system receives the audit log and validates its structure and required fields.
3. The system identifies sensitive PII fields in the audit log.
4. The system masks detected PII values such as credit card numbers and Social Security Numbers.
5. The system creates the sanitized audit log.
6. The system generates a SHA-256 integrity hash for the sanitized log.
7. The system stores the sanitized audit log together with its integrity hash.
8. The system records successful processing of the audit log.
9. The sanitized audit record becomes available for authorized searching, filtering, and auditing.

Alternate Flows**A1 — Invalid JSON or Missing Required Fields**

1. The system receives a malformed or incomplete audit log.
2. JSON or required-field validation fails.
3. The system rejects the audit log.
4. The validation error is recorded.
5. The unsanitized log is not stored.

A2 — PII Detected

1. The system identifies one or more sensitive PII fields.
2. The system replaces each detected sensitive value with a masked representation.
3. Processing continues using the sanitized audit log.
4. The sanitized log is hashed and stored.

A3 — Hashing or Storage Failure

1. PII masking completes successfully.
2. SHA-256 hash generation or storage fails.
3. The system does not mark the audit log as successfully processed.
4. The failure is recorded for investigation or retry.
5. No incomplete or unverifiable record is presented as successfully stored.